

LC 2028: Find Missing Observations

Detailed Problem Breakdown and Solution

Bloomberg

Problem Understanding

- You have a 6-sided dice (values 1 to 6).
- n observations went missing.
- The average of $n + m$ rolls is given.
- **Input:** The mean and an array of m known rolls.
- **Output:** A vector of n missing rolls such that the overall mean remains consistent.

Example Walkthrough

Given: `rolls = [3, 2, 4]`, `mean = 4`, `n = 2`

(S1) Total dice count = $(3 + 2) = 5$ dice

(S2) Total sum required = $(5 \times 4) = 20$

(S3) Current sum from known rolls = $(3 + 2 + 4) = 9$

(S4) Missing dice sum = $(20 - 9) = 11$

Question: Can you write 11 as a sum of 2 valid dice rolls? **YES (5 + 6)**

Greedy Observation

Distribute the required sum among the missing n dice by assigning the minimum value first, then greedily distributing the remainder.

Initially give every die 1:

[1, 1] remaining sum = $11 - 2 = 9$

Add up to 5 (max possible to reach 6) to the first die:

+5

[6, 1] remaining sum = $9 - 5 = 4$

Add remaining to the next die:

+4

[6, 5] remaining sum = 0

Result: return [6, 5]

C++ Implementation

```
class Solution {
public:
    vector<int> missingRolls(vector<int>& rolls, int mean, int n) {
        int m = rolls.size();
        int currrsum = accumulate(rolls.begin(), rolls.end(), 0);
        int totsum = mean * (n + m);

        int rem = totsum - currrsum;

        // Cause of range of dice
        if (rem < n || rem > 6 * n) {
            return {};
        }

        // Sab dice ko sabse pehle minimum possible sum denge
        vector<int> ans(n, 1);
        rem = rem - n; // Sab ko 1 1 de diya

        for (int i = 0; i < n && rem > 0; i++) {
            ans[i] += min(5, rem);
            // -1 cause maine sirf 5 ya rem ko redistribute kiya he,
            // 1 to already tha dice me
            rem -= ans[i] - 1;
        }

        return ans;
    }
};
```